

MUHAMMAD SAIF

Machine Learning Engineer | Python Developer

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PROFESSIONAL SUMMARY

Machine Learning Engineer & Python Developer with 5+ internships building CNN/LSTM models and production Python backends — including a Top 4 finish at Pakistan Innovative 2026 among 1,000+ competing projects, currently incubated at NIC Faisalabad.

Built and deployed production-oriented CNN models achieving 90%+ OCR accuracy and ~85% mental health classification accuracy, served via FastAPI REST endpoints with real-time monitoring dashboards. Seeking a Machine Learning Engineer or Python Developer role to build, train, and ship models and backend systems that reach real users and deliver measurable impact.

TECHNICAL SKILLS

AI & ML: TensorFlow, Keras, Scikit-learn, CNN, LSTM, YOLO, Transfer Learning, Model Deployment, Hyperparameter Tuning, Cross-Validation, Computer Vision, NLP, OCR, Deep Learning

Programming: Python, SQL, C++, Java, JavaScript

Backend: FastAPI, Flask, REST APIs, Docker, MySQL, Supabase, Firebase, PHP

Libraries: OpenCV, NumPy, Pandas, Matplotlib, Seaborn, PyMuPDF, Tesseract OCR, python-docx, face_recognition

Automation: Make.com, n8n, Prompt Engineering, LLM Integration (OpenAI / Anthropic APIs), API Webhooks, Workflow Automation

Tools & Frontend: Git, GitHub, Postman, VS Code, PyCharm, pytest, HTML5, CSS3, Bootstrap, Tailwind CSS

PROFESSIONAL EXPERIENCE

AI Automation Intern | *Dot Tribe (Remote)*

Oct 2025 – Dec 2025

- Reduced client workflow processing time by automating 3 recurring business processes using Make.com, cutting an estimated 5+ hours/week of manual effort per workflow.
- Accelerated integration delivery by building a reusable webhook + API connection framework, enabling same-day deployment of AI services — including LLM-powered endpoints — into business applications.
- Improved scenario reliability by designing structured test protocols, achieving zero-failure runs across all production automations before client handoff.

Machine Learning Intern | *Gamica Cloud, Faisalabad*

Jun 2025 – Jul 2025

- Increased OCR text-extraction accuracy to 90%+ by engineering a Tesseract OCR + OpenCV preprocessing pipeline, eliminating manual correction for downstream document workflows.
- Delivered a production-ready CNN gender-recognition model in TensorFlow/Keras — validated with cross-validation, benchmarked with Scikit-learn metrics, and served as a real-time inference endpoint.
- Cut data pipeline build time by constructing reusable preprocessing modules adopted across 2 internal computer vision projects.

Python Developer Intern | *Compiling Logics, Faisalabad*

May 2025 – May 2026

- Eliminated an estimated 3+ hours/day of repetitive data processing by building Python automation scripts with OOP design patterns, freeing team capacity for higher-priority development tasks.
- Improved sprint delivery speed by identifying and resolving 4 coding-workflow bottlenecks, reducing backend turnaround time across the team.
- Integrated REST APIs and SQL/Supabase databases into backend services, applying pytest-based unit testing to maintain code reliability across iterative builds.

Web Development Intern | *Career Institute (Onsite)*

Jun 2024 – Jul 2024

- Delivered 3 responsive websites using HTML, CSS, JavaScript, Bootstrap, and PHP — tested across 5+ device breakpoints to ensure consistent UI on mobile and desktop.
- Reduced frontend rework by building reusable component templates, cutting estimated per-page development time by 30%.

Additional Experience:

Social Media Marketing Intern — Digital Empowerment Network (DEN)

Oct 2023 – Feb 2024

General Secretary — Entrepreneurship Society, The University of Faisalabad

Dec 2024 – Jun 2025

FEATURED PROJECTS

Multi-Sensor Mental Health Detection System — Final Year Project | Top 4, Pakistan Innovative 2026 — Python, TensorFlow, OpenCV, FastAPI, EEG, EDA/GSR, PPG, MySQL

- Achieved ~85% classification accuracy on a multimodal mental health detection system by fusing EEG, EDA/GSR, PPG, and facial emotion recognition data using CNN + LSTM deep learning models.
- Deployed the full system via FastAPI as a REST endpoint, serving a real-time monitoring dashboard with trend analysis and personalized mental health recommendations.
- Recognized as Top 4 Finalist at Pakistan Innovative 2026 (1,000+ projects, IEEE-backed); currently incubated at NIC Faisalabad as an active startup venture.

Facial Recognition System — Python, OpenCV, TensorFlow, Keras, face_recognition, NumPy, FastAPI

- Achieved high-accuracy real-time identity matching by implementing CNN-based face embedding pipelines, handling multiple simultaneous faces across varying lighting conditions and angles.
- Reduced false-positive identity matches by integrating face encoding comparison logic with a persistent local database, enabling reliable attendance and access-control use cases.

- Deployed as a REST endpoint via FastAPI, enabling integration with third-party access management systems.

Gender Detection System — Python, TensorFlow, Keras, OpenCV, Scikit-learn, NumPy

- Trained a CNN binary classification model for real-time facial gender prediction on a large-scale dataset with data augmentation, achieving production-ready inference performance.
- Benchmarked model performance using Scikit-learn metrics (precision, recall, F1) and deployed with OpenCV for real-time webcam integration with bounding-box overlays.

AI Calling Agent — Python, OpenAI API, Make.com, Prompt Engineering, CRM Integration

- Built an LLM-powered AI calling assistant using prompt engineering and OpenAI API integration, automating lead qualification and appointment scheduling workflows connected to CRM systems via Make.com.

Python Tools & Utilities Suite — PyMuPDF, PyPDF2, python-docx, WeasyPrint, Tesseract OCR, OpenCV

- Delivered a comprehensive document manipulation toolkit — PDF editor/compressor, split/unlock/protect security module, digital signing, and format conversions (Word/PPT to PDF, HTML to PDF, PDF to Image).
- Integrated Tesseract OCR + OpenCV for scanned document extraction and NLP-based text summarization, processing documents end-to-end without manual intervention.

Slip OCR Extraction System — Python, OpenCV, Tesseract OCR, NLP, FastAPI

- Engineered an automated receipt-processing pipeline — extracted structured data from unstructured scanned documents using Tesseract OCR + NLP preprocessing, served via FastAPI for downstream integration.

Restaurant POS System | Coffee Shop Mobile App

- Restaurant POS: Python/SQL/Supabase platform with billing, inventory, kitchen management, and reporting modules.
- Coffee Shop Android App: Java/XML/Firebase with authentication, real-time database, and ordering features.

EDUCATION

Bachelor of Science in Computer Science (BSCS) — The University of Faisalabad

CGPA: 3.22 / 4.00 | Specialization: Machine Learning & Data Science

CERTIFICATIONS

- Machine Learning Specialization — NAVTTC x Gamica Cloud (A+ Grade) | AI: Machine Learning, Deep Learning & Communication | Mar–Jun 2025
- Data Science with Python — Gamica Cloud | Feb 2024
- Data Analytics and Visualization Job Simulation — Accenture x Forage | Aug 2024
- Data Visualisation: Empowering Business with Effective Insights — Tata x Forage | Sep 2024
- Nestle E-Learning 2024: Sustainability — Nestle Connect

RECOGNITION & IMPACT

- Top 4 Finalist — Pakistan Innovative 2026 (1,000+ competing projects, IEEE-backed competition, May 2026)
- NIC Faisalabad Incubatee — Active startup founder at the National Incubation Center Faisalabad
- Hult Prize On-Campus Runner-Up — Led social entrepreneurship startup team, The University of Faisalabad (Feb 2024)
- TEDxTUF Organizer — Successfully organized TEDxTUF Unlocking Tomorrow at The University of Faisalabad (Nov 2025)
- Congress Secretary — 10th Academic Studies Congress ASC-2025/Fall, National Business School (Oct 2025)
- Organising Team — International Seminar on Sustainable Development, TUF (Feb 2026)
- Organising Team — Two-Day International Workshop on Ethics & AI for Improving Quality Education in HEIs (Feb 2026)
- Participant — 3rd International Conference on Advance STEAM Education (ICASE-2025), Nov 2025
- SME Launchpad Session — Certificate of Participation, State Bank of Pakistan (Feb 2026)

LANGUAGES & INTERESTS

Languages: English (Professional) | Urdu (Native)

Interests: Artificial Intelligence, Machine Learning Research, Computer Vision, Workflow Automation, Chess, Rubik's Cube